

# Qatar-2b

R-1660

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_Qatar-2\_190319 · created 2026-08-02T09:23:47+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

## Results

|                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| Mid-Transit Time (Tmid)                           | 2458561.91341 +/- 0.00078 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.226 +/- 0.066                   |
| Transit depth (Rp/Rs)^2                           | 5.12 +/- 3.0 [%]                  |
| Orbital Inclination (inc)                         | 89.61 +/- 1.72                    |
| Airmass coefficient 1 (a1)                        | 3.8 +/- 0.88                      |
| Airmass coefficient 2 (a2)                        | -0.94 +/- 0.75                    |
| Scatter in the residuals of the lightcurve fit is | 24.2 %                            |
| Best Comparison Star                              | #3 - [455, 189]                   |
| Optimal Aperture                                  | 5.59                              |
| Optimal Annulus                                   | 9.32                              |
| Transit Duration (day)                            | 0.0803 +/- 0.0048                 |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                            |
|---------------------------|--------------------------------------------|
| Rp/R■ fit vs published    | 0.226 ± 0.066 vs 0.16327 (deviation 0.95σ) |
| Duration fit vs published | 0.0803 d vs 0.0767 d (deviation 0.76σ)     |
| Mid-time O–C              | 1.8 min                                    |
| Depth SNR                 | 3.4                                        |
| Residual scatter          | 24.2 %                                     |

## Light curve

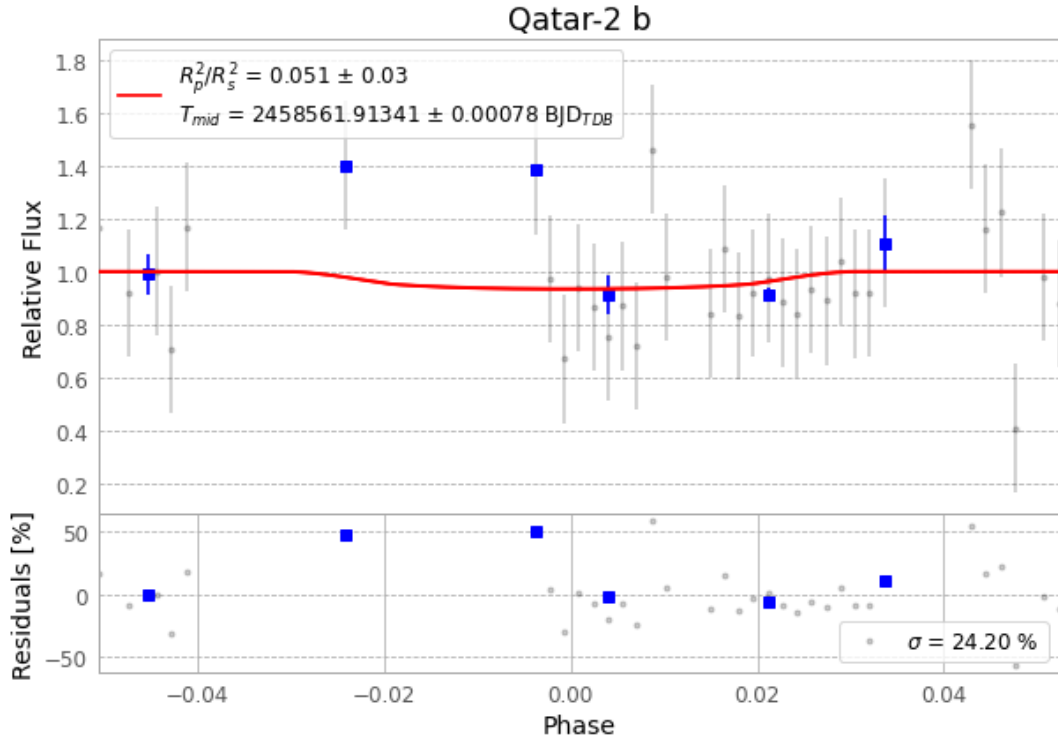


Figure 1 — FinalLightCurve\_Qatar-2 b\_2019-03-19.png (SHA-256 c6cad5dae6d0714a...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [470, 122], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 ceda442402fb392f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

72 FITS frames (47 MB), Qatar-2190319080915.FITS → Qatar-2190319114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-190319.log (bb793a3f2dd4...), FinalLightCurve\_Qatar-2 b\_2019-03-19.png (c6cad5dae6d0...), FinalLightCurve\_Qatar-2 b\_2019-03-19.csv (cf9ea0d5fb7f...)

## Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 09:23Z.